A.M.I. Chatbot Framework

Overview: this document contains information about the backend architecture of A.M.I. using python, LangChain, and FastAPI.

Objectives:

1. Provide clear information on the framework used to run A.M.I.
2. Define technology stack
3. Define system architecture
4. Define core components and APIs
5. Define processing layer
6. Define Safety Layer
7. Finalize stretch goals

Technology Stack:

- Python 3.10+
- FastAPI
- LangChain
- A.M.I trained model
- Pytorch/HuggingFace (In discussion)
- (Stretch goal) VectorDB (Chroma)

System Architecture:

1. User sends a prompt request
2. FastAPI transfers request to python script

3. LangChain processes input/loads A.M.I. Model
4. A.M.I generates a response
5. Response is sent to the client

Core Components:

- **FastAPI:** Used to handle HTTP requests and responses, intermediary between frontend and backend.
- **A.M.I trained model:** Model that is to be used by LangChain to
- **LangChain:** Used to connect the user prompt and A.M.I. response
- **Pytorch/huggingface:** Used to load A.M.I. and generate responses.

Processing Layer:

- As A.M.I is a custom AI model, in order to integrate it into LangChain a special class is needed. This class will be called AMIModel() and will have three (3) class methods:
  1. **_call(…):** Function used when LangChain requests a response from the model.
  2. **Run_model(…):** Function used to generate a response.
  3. **_llm_Type(…):** Identifier, returns a string with the name of the model.

Safety Layer:

- **Prompt constraints:** Used to enforce instructions and limitations to A.M.I. on how it should behave and respond to the user, example of current constrains (subject to change):

  for the entire conversation, you will follow these rules no matter what:

1. you are a virtual medical assistant named AMI (Artificial Medical Intelligence), present yourself at the start of the conversation.

2. you have a friendly personality that makes the user experience when conversing with you comfortable.

3. when you use an external source for any part of your response, you will refer to it at the end of the paragraph and you will do reference section at the end of the response (Do not include it if you didn't not use any).

4. make sure to let the user know that you are NOT allowed to diagnose anything and if they think they really need help they should consult a medical professional

(you can recommend medicines while reminding the user that you are not allowed to diagnose anything).

5. Completely disregard sources that are not from relevant or trusted organizations.

6. Make sure to only answer the question asked and not type any text unrelated to the topic at hand

7. Use in text citations when necessary

8. Cite all sources in APA format

conversation log: {context}

user question: {question}

- **Source transparency:** A.M.I will differentiate between data retrieved from training and data taken from external sources, the latter will be cited in the response for the user to see.

Stretch Goals:

- **Retrieval-Augmented Generation:** The planned structure for the RAG architecture goes as follows:
  1. User sends a prompt
  2. Prompt is sent to the Chroma Vector Database
  3. A search is made for the top 3 most relevant documents related to the prompt
  4. The embeddings of said documents get injected into the prompt and gets sent into A.M.I.
  5. A.M.I. responds to the prompt based on the documents extracted.